RHCSA 10 (EX200) Mastery Guide

RHCSA 10 (EX200) Complete Certification Guide *Master Red Hat Enterprise Linux 10 Administration (Fully Updated for the RHCSA 10 Exam Objectives)*

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1. Introduction

The Red Hat Certified System Administrator (RHCSA) certification is one of the most recognized credentials in Linux system administration. With **RHEL 10**, Red Hat has updated the objectives to emphasize modern automation and storage while deprecating older technologies.

This guide is designed to help you:

- Understand the full scope of the EX200 exam.
- Recognize performance-based task expectations.
- Review all objectives with hands-on labs.
- Build a strategic, focused study plan.

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2. What's New / Changed in RHCSA 10

Removed from the Exam

- ❌ **Podman & Containers:** Removed entirely.
- ❌ **Stratis:** Layered storage management is no longer tested.
- ❌ **Module Streams:** `dnf module` is deprecated.
- ❌ **Set-GID for Collaboration:** Removed as a specific standalone objective.

Added or Updated in the Exam

- ✓ **Flatpak Application Management:** Configure remotes and manage desktop apps.
- ✓ **Systemd Timers:** Preferred over Cron for scheduling.
- ✓ **Modern Storage (VDO via LVM):** VDO is now a native LVM type (`lvcreate --type vdo`).

Unchanged but Essential

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- ✓ Systemd service management
- ✓ LVM administration & swap
- ✓ SELinux context management
- ✓ NFS and Autofs
- ✓ Bootloader (grubby)
- ✓ FirewallD rich rules
- ✓ User and group management

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3. Exam Format & Expectations

Format:

- **Type:** Performance-based (Hands-on only).
- **Duration:** ~2.5 hours.
- **Environment:** RHEL 10 virtual machines.

Grading Criteria:

1. **Functionality:** Does it work?
2. **Persistence:** Does it survive a reboot?
3. **Accuracy:** Are paths, names, and permissions exact?

Common Mistakes:

- ❌ Forgetting `--permanent` in firewall rules.
- ❌ Editing `/proc/sys` instead of `/etc/sysctl.d/`.
- ❌ Not enabling services (`systemctl enable`).
- ❌ Incorrect SELinux contexts preventing service startup.

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4. Setting Up Your Study Environment

Requirements:

- **Hypervisor:** VirtualBox, VMware, or KVM.
- **ISO:** RHEL 10 (or AlmaLinux/Rocky Linux 10).
- **Specs:** 4 GB RAM, 2 vCPUs, 40 GB Disk per VM.

Lab Architecture:

- **ServerA (Infrastructure):** 192.168.1.100 (Repo/NFS/SSH server).

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- **ServerB (Exam Node):** 192.168.1.11 (Your practice machine).

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5. RHCSA 10 Objectives & Core Skills

Domain I: Essential Tools

Key Concepts: File operations, grep/sed, archiving.

Hands-On Lab: Archiving and Searching

1. Create `/archive`.
2. Find files in `/etc` ending in `.conf` > 2KB.
3. Copy them to `/archive`.
4. Create `/root/configs.tar.gz` containing `/archive`.

Verification:

- [ ] `ls -lh /root/configs.tar.gz` (Check size/existence)
- [ ] `tar -tf /root/configs.tar.gz` (Verify contents)

Domain II: Managing Software

Key Concepts: DNF, Groups, Flatpak.

Hands-On Lab: Flatpak Deployment

1. Add Flathub remote: `flatpak remote-add --if-not-exists flathub <url>`.
2. Install `org.gnome.Calculator`.
3. Install "Development Tools" group via DNF.

Verification:

- [ ] `flatpak list` (Shows Calculator?)
- [ ] `dnf group list installed` (Shows Dev Tools?)

Domain III: Shell Scripting

Key Concepts: Bash shebang, loops, arguments.

Hands-On Lab: User Creator Script

1. Create `/usr/local/bin/create_users.sh`.
2. Loop through arguments (`$@`) to create users.

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3. Make executable (`chmod +x`).

Verification:

- [ ] `./create_users.sh u1 u2` -> `id u1` (User exists?)

Domain IV: Operating Running Systems

Key Concepts: Boot targets, Tuned, Grubby.

Hands-On Lab: Tuning

1. Set default target to `multi-user.target`.
2. Set Tuned profile to `virtual-guest`.

Verification:

- [ ] `systemctl get-default` -> `multi-user.target`
- [ ] `tuned-adm active` -> `virtual-guest`

Domain V: Storage Administration

Key Concepts: LVM, VDO, Swap.

Hands-On Lab: VDO Storage

1. Create VDO LV `myvdo` (VG `vdovg`, Disk `sdb`).
2. Physical: **5GB**, Virtual: **50GB**.
3. Format XFS, mount at `/vdo_data` (Persistent).

Verification:

- [ ] `lvs -o+vdo_compression` (Enabled?)
- [ ] `df -h /vdo_data` (Shows ~50GB?)
- [ ] **Edge Case:** Did you install `vdo` package? If not, `lvcreate` fails.

Domain VI: Filesystems & NFS

Key Concepts: AutoFS, Mounting.

Hands-On Lab: AutoFS

1. Install `autofs`.
2. Map `/data/public` to `servera:/shares/public`.

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Verification:

- [ ] `ls /data/public` (Files appear?)
- [ ] `mount | grep nfs` (Mounts on demand?)

Domain VII: Systemd Services & Timers

Key Concepts: Timers vs Cron.

Hands-On Lab: Daily Job

1. Create script `/opt/job.sh`.
2. Create `job.service` and `job.timer` (Daily @ 23:00).
3. Start timer.

Verification:

- [ ] `systemctl list-timers` (Shows next run?)

Domain VIII: Networking

Key Concepts: NetworkManager, Static IPs.

Hands-On Lab: Static IP Configuration

1. Create connection `lab-static` on interface `enp0s3`.
2. **IPv4:** `192.168.50.50/24`. **GW:** `192.168.50.1`. **DNS:** `8.8.8.8`.
3. Set hostname to `serverb.lab.example.com`.
4. Activate connection.

Verification Checklist:

- [ ] `nmcli con show` (Is `lab-static` green/active?)
- [ ] `ip addr show enp0s3` (Expected: `inet 192.168.50.50/24`)
- [ ] `hostnamectl` (Static hostname correct?)
- [ ] **Edge Case:** If IP doesn't change, did you run `nmcli con up lab-static`?

Domain IX: Users & Groups

Key Concepts: ACLs, SGID, Collaboration.

Hands-On Lab: Collaboration Directory

1. Create group `finance`. Add user `sarah`.
2. Create `/finance_data`. Group owner: `finance`.

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3. Permissions: `2770` (SGID + RWX for group).
4. **ACL:** User `bob` (not in finance) needs Read-only access.

Verification Checklist:

- [ ] `ls -ld /finance_data` -> `drwxrws---` (SGID bit `s` present?)
- [ ] `getfacl /finance_data` -> `user:bob:r-x`
- [ ] **Test:** `su - sarah` -> `touch /finance_data/file` (Should succeed).
- [ ] **Edge Case:** `su - bob` -> `touch /finance_data/file` (Should fail: Permission Denied).

Domain X: Security (Firewall & SSH)

Key Concepts: SSH Hardening, Rich Rules.

Hands-On Lab: SSH & Firewall

1. Generate SSH keys (`ssh-keygen`). Copy to localhost.
2. Edit `/etc/ssh/sshd_config`: `PasswordAuthentication no`.
3. **Firewall:** Block traffic from `10.0.0.5` using a Rich Rule.

Verification Checklist:

- [ ] `ssh localhost` (Login without password?)
- [ ] `ssh -o PreferredAuthentications=password localhost` (Should fail: "Permission denied (publickey)").
- [ ] `firewall-cmd --list-rich-rules` (Shows `rule ... source address="10.0.0.5" reject`?)
- [ ] **Edge Case:** Did you restart `sshd` and reload `firewalld`?

Domain XI: SELinux

Key Concepts: Contexts, Ports, Booleans.

Hands-On Lab: Web Server on Custom Port 82

1. Change Apache port to `82` in `/etc/httpd/conf/httpd.conf`.
2. **SELinux:** Add port 82 to `http_port_t`.
3. **Firewall:** Open port 82/tcp.
4. Start `httpd`.

Verification Checklist:

- [ ] `semanage port -l | grep 82` (Shows `http_port_t`?)
- [ ] `systemctl is-active httpd` (Active?)

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- [ ] `curl localhost:82` (Returns page or 403?)
- [ ] **Edge Case:** If `curl` fails, check if you created `index.html` and ran `restorecon -Rv /var/www/html`.

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6. Exam Strategy Checklist

- ✓ **Read instructions carefully:** Hostnames, paths, and usernames are graded strictly.
- ✓ **Reboot before submitting:** If it doesn't work after a reboot, you get zero points.
- ✓ **Validate constantly:** Use `systemctl status`, `ls -Z`, and `journalctl -xe`.
- ✓ **Stay calm:** Use `rd.break` (Rescue Mode) if you lock yourself out.

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